PROJECT REPORT

SMARTPHONE HOLDER / STAND FOR LAPTOP

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CONTENTS

Sr. No		Name of Topic	Page No.
1		Introduction of Mechanical Prototyping	5
	1.1	Advantages	5
	1.2	Limitations	6
	1.3	Applications	6
	1.4	Techniques of Rapid Prototyping	7
2		Concept Presentation	8
3		CAD Modeling	9
	3.1	Project	9
4		Methodology	11
	4.1	Pre-Processing	12
	4.2	Post-Processing	13
5		Different Materials of 3D Printing Machines	13
6		Applications of Project and Group Assembly	14
7		Bill of Material	14
8		Conclusion	15
		References	

ABSTRACT

Our product is **"Smartphone Holder/Stand for Laptop."** The smartphone holder is designed as an innovative accessory that allows users to conveniently attach their mobile phones to a laptop screen. This enhances multitasking, enabling users to take video calls, follow tutorials, or monitor notifications while simultaneously working on their laptop.

The prototype aims to explore the design and functionality of a compact, lightweight, and durable holder using **rapid prototyping techniques**. The model was designed on **Free CAD** and fabricated using **3D printing technology**, which provides flexibility, accuracy, and fast prototyping.

This project demonstrates how additive manufacturing can be effectively applied to create useful daily-life accessories that improve productivity and ergonomics.

ACKNOWLEDGEMENT

We would like to express our special thanks of gratitude to the **Mechanical Prototyping Faculty**, who gave us the opportunity to work on this innovative project and guided us in understanding the fundamentals of rapid prototyping.

We sincerely thank our project guide, **Mr. Ashwin S. Chandore**, for his constant support, valuable guidance, and encouragement throughout the development of our idea. His insightful feedback and expertise helped us refine our design and overcome challenges at every stage.

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1. Introduction of Mechanical Prototyping

Mechanical prototyping is the process of transforming an idea or concept into a physical model that can be tested, analyzed, and improved before moving into large-scale production. It allows designers and engineers to visualize the product, check its functionality, and identify possible improvements at an early stage.

In today's competitive world, rapid prototyping has become an essential tool for product development. With the help of 3D modeling software and advanced manufacturing techniques, prototypes can be created quickly, accurately, and at a lower cost compared to traditional methods. This helps in reducing design errors and saves both time and resources.

Mechanical prototyping also provides a platform for experimentation, enabling students and professionals to convert innovative ideas into real products. By using additive manufacturing techniques like 3D printing, even complex shapes and customized designs can be fabricated with ease.

Thus, mechanical prototyping not only supports learning and innovation but also bridges the gap between theoretical knowledge and practical application.

1.1 Advantages :

1. **Faster product development:** Prototyping allows designs to be created and tested quickly, which reduces the overall development time.
2. **Cost saving:** Errors can be detected in the prototype stage, which prevents unnecessary expenses in the final production.
3. **Clear product understanding:** Physical models help to understand the actual size, shape, and usability of the product better than drawings.
4. **Easy customization:** Prototypes can be adjusted and modified to meet specific user requirements.
5. **Encourages innovation:** It provides freedom to try new ideas and experiment with creative designs.
6. **Practical testing:** The product can be tested in real conditions to check if it works as expected.

1.2 Limitations

1. **Material strength:** Prototypes made by 3D printing or other rapid prototyping methods are often weaker than actual production materials.
2. **High equipment cost:** Advanced prototyping machines and software can be expensive for students or small projects.
3. **Surface quality issues:** The finish of prototypes may not always be smooth, and extra processing may be needed.
4. **Time for large models:** Creating big or complex designs can take a long time to print or process.
5. **Not suitable for final use:** Most prototypes are only for testing and cannot always be used as the final product.

1.3 Applications

1. **Testing product concepts:** Prototyping helps to check whether a new design idea is practical and works as expected.
2. **Education and research:** Engineering students and researchers use prototypes to learn concepts and test their projects in real life.
3. **Medical field:** Doctors use prototypes for making custom prosthetics, implants, and surgical models.
4. **Automotive and aerospace:** Prototypes are used for designing lightweight parts and testing them before production.
5. **Consumer products:** Useful for developing everyday items and accessories, such as holders, stands, or household tools.

1.4 Techniques of Rapid Prototyping

1. 3D Printing (Additive Manufacturing):

This method builds the product layer by layer directly from a digital model. It is widely used because it is simple, cost-effective, and suitable for making complex shapes in less time.

2. CNC Machining:

A subtractive process where material is cut away from a solid block using computer-controlled machines. It provides high strength and accuracy but requires more material and time compared to 3D printing.

3. Stereolithography (SLA):

SLA uses liquid resin that is hardened by a UV laser to form detailed and accurate prototypes. It is mainly used for models requiring smooth surfaces and fine details.

4. Selective Laser Sintering (SLS):

In this technique, a laser beam fuses powdered material layer by layer to form the object. It is suitable for strong, durable parts and works with plastics as well as metals.

5. Fused Deposition Modeling (FDM):

This process melts a thermoplastic filament and deposits it layer by layer to create the product. It is the most common type of 3D printing, ideal for educational, research, and small-scale applications.

2. Concept Presentation

Problem Statement

In today's digital world, people often need to use their smartphones alongside their laptops for multitasking—such as attending video calls, following tutorials, checking notifications, or using mobile apps while working on the computer. However, placing the phone flat on the table or holding it by hand is uncomfortable, reduces productivity, and affects posture. There is a need for a simple, compact, and effective accessory that allows the smartphone to be securely placed near the laptop screen for easy access and better usability.

Solution

To overcome this issue, we designed and developed an innovative Smartphone Holder/Stand for Laptop. The holder can be easily attached to the side of the laptop screen and securely holds the smartphone in a convenient position. This makes multitasking easier, improves workspace organization, and saves time for the user.

Our prototype is lightweight, portable, and made using 3D printing technology, which ensures durability and easy customization. The design allows for quick attachment and removal, making it user-friendly for students, professionals, and anyone who frequently uses both devices together.

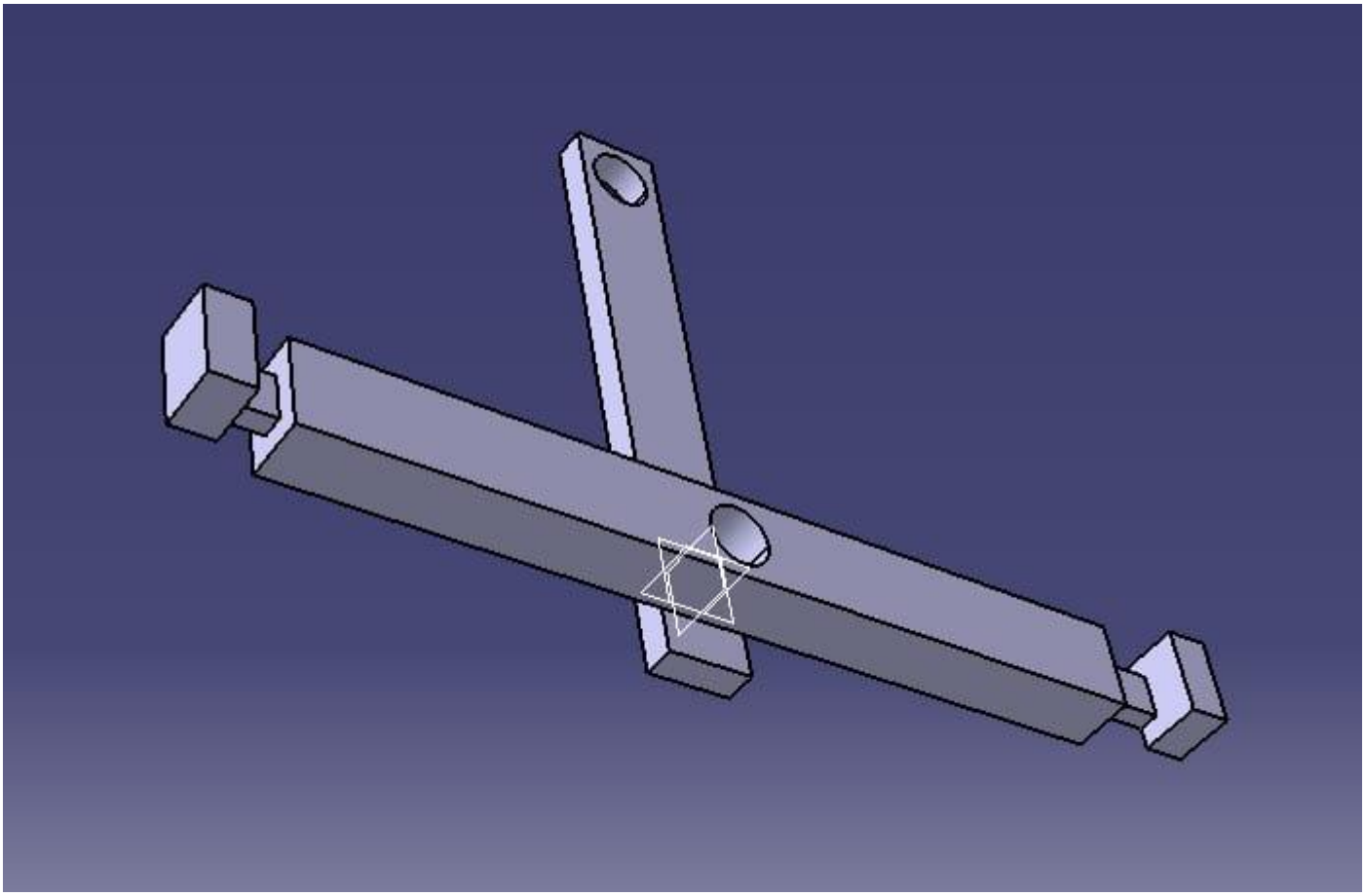
3. CAD Modeling

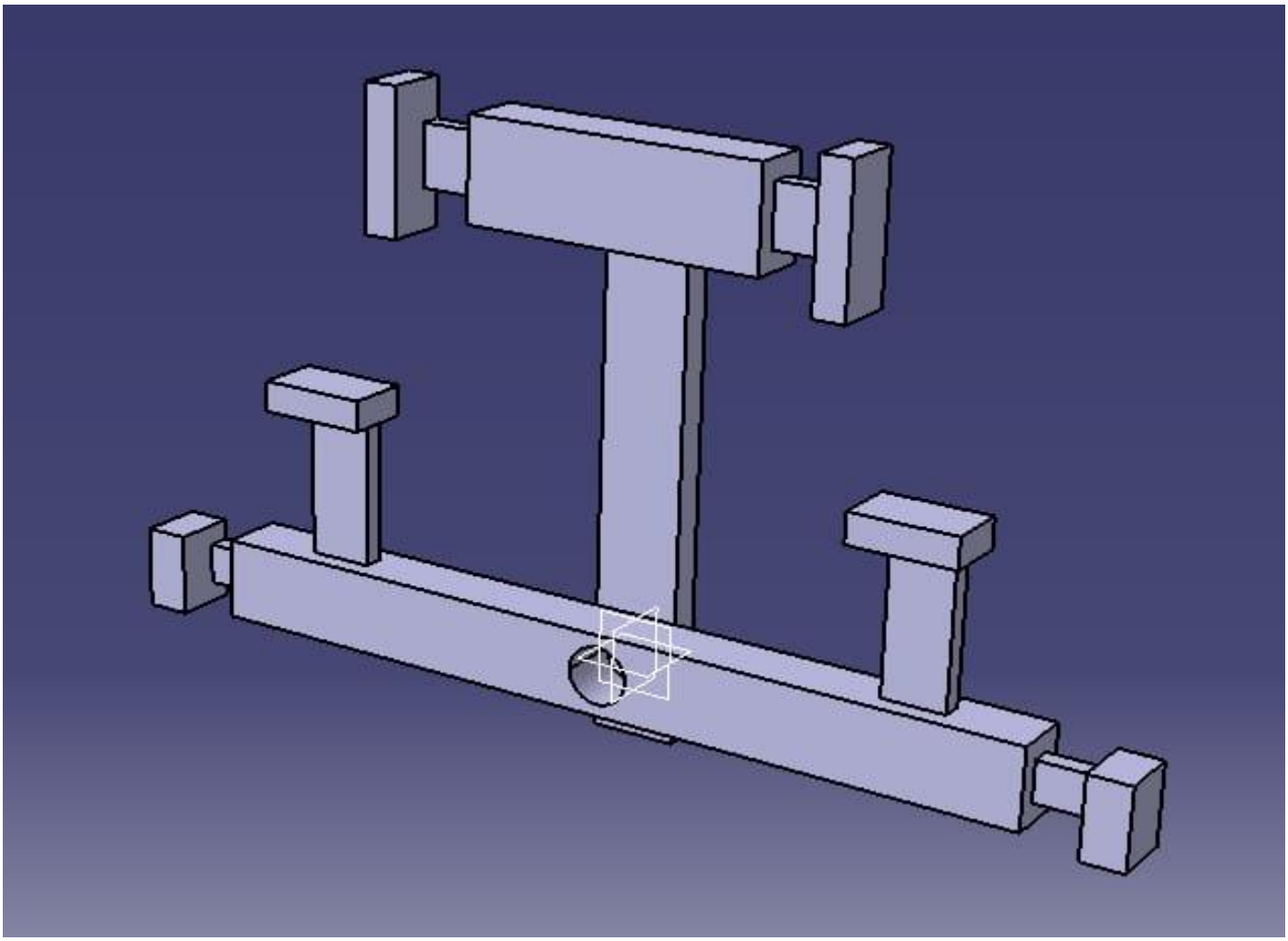
Project

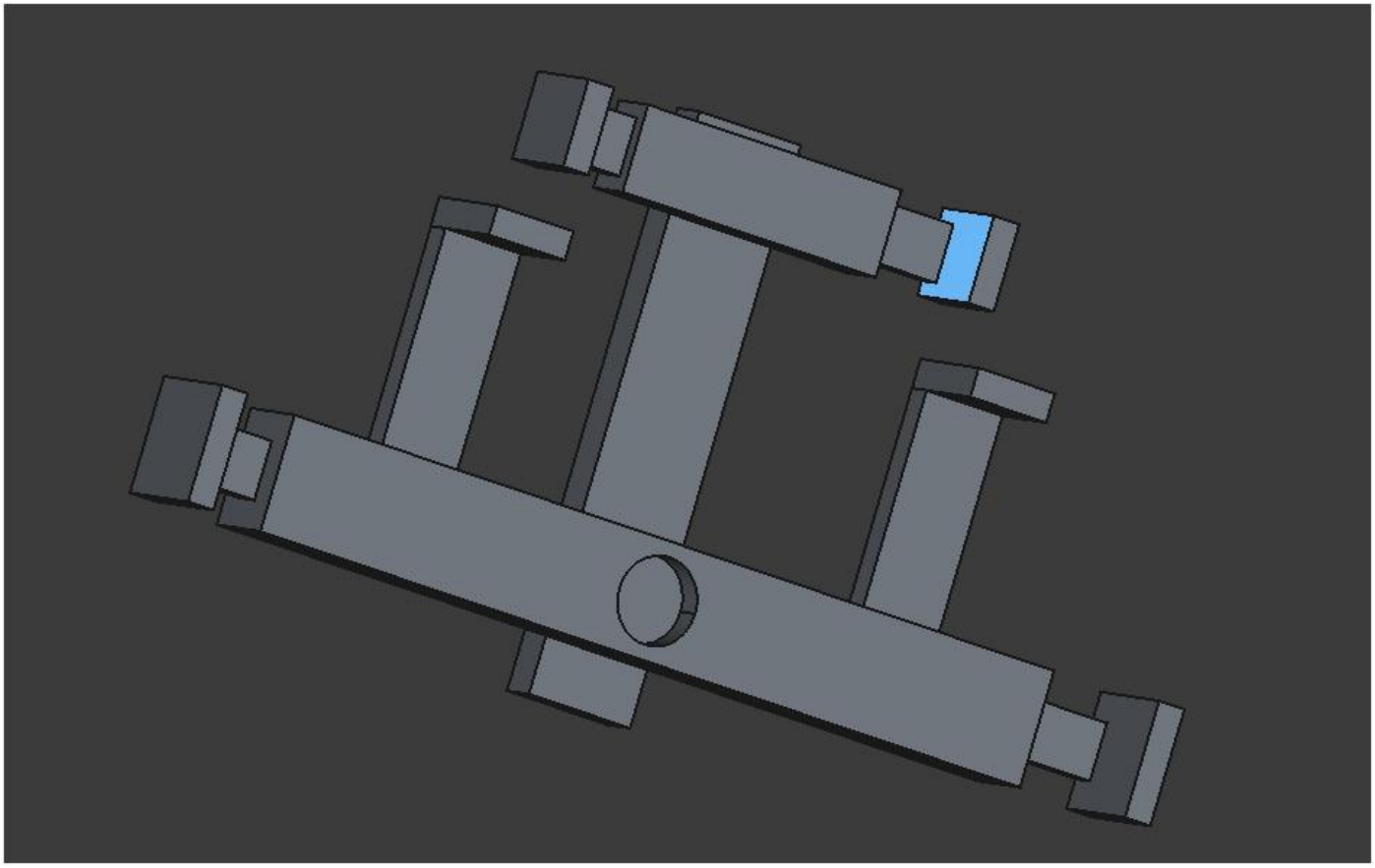
The **Smartphone Holder/Stand for Laptop** was designed using **CATIA/ Free CAD**. CAD software allows us to create accurate 3D models, visualize the final product, and make modifications easily before manufacturing

Process:

- **Basic Shape Creation:** The first step was to design the overall shape of the holder in CAD software, keeping it simple and functional.
- **Dimension Adjustment:** The model was refined by adjusting the dimensions to ensure proper balance, stability, and a compact design.
- **Adding Features:** Additional features were included in the model to improve usability and make the product suitable for daily use.
- **View Generation:** Different views such as front, side, and isometric were generated in CAD to study the product from multiple angles.
- **Finalization:** The completed 3D model was finalized and exported for slicing, which prepared it for 3D printing.







4. Methodology

A) CAD Software used:

We used **FreeCAD/CATIA** for designing the smartphone holder/stand. This software helped us to create accurate 3D models, check proportions, and make improvements before moving to printing.

B) Slicing Software used:

The CAD model was processed using **Ultimaker Cura**, which converted the design into layers and generated the G-code for 3D printing. Parameters such as infill, layer height, and print speed were set in this stage.

C) 3D Printer used:

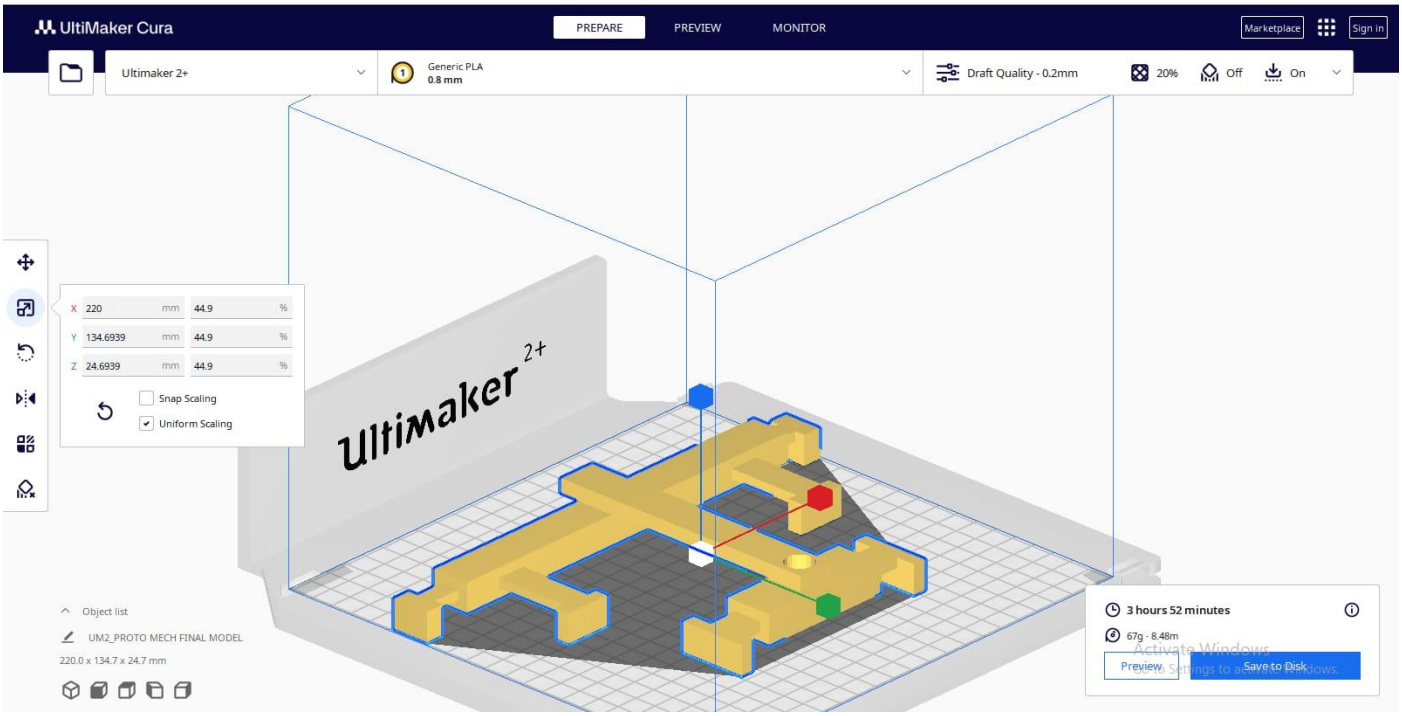
The model was printed on an **FDM-based 3D printer** (e.g., Creality Ender or Prusa). This printer works by depositing melted thermoplastic filament layer by layer to form the final object.

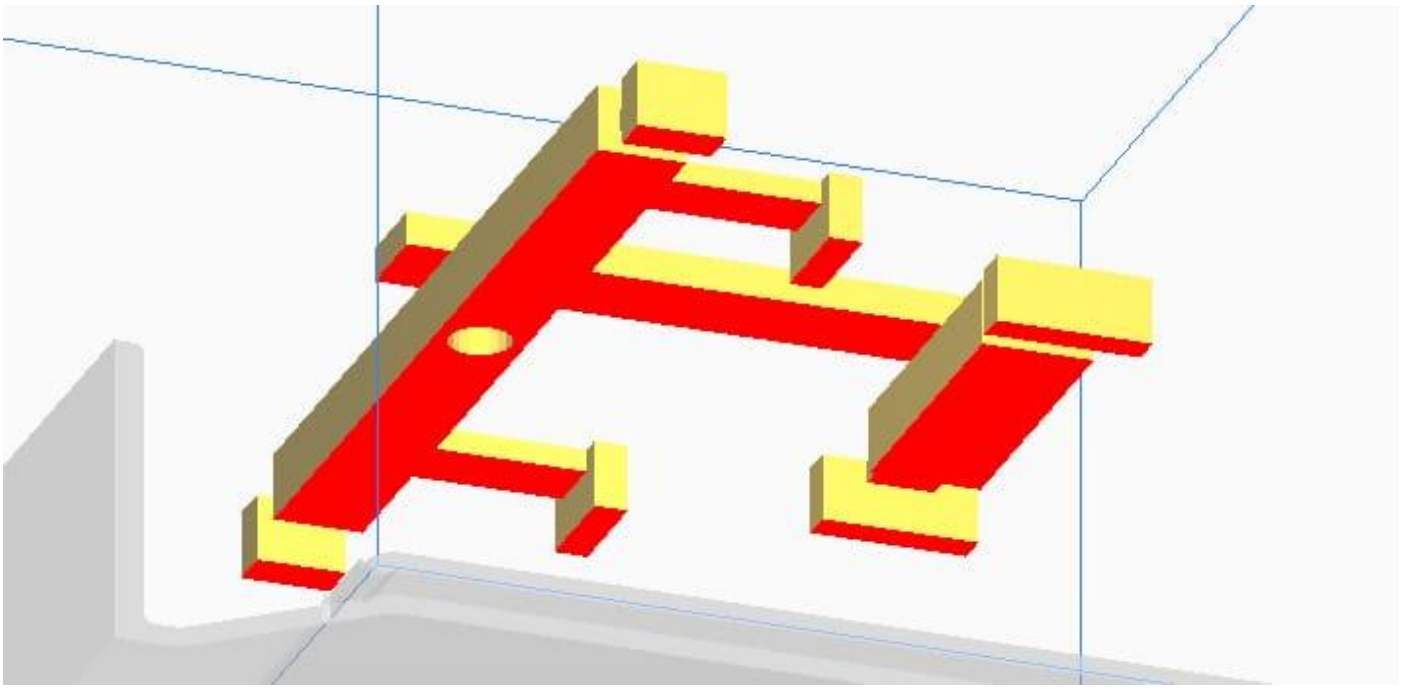
D) Rapid Prototyping Techniques/Method Used:

We used the **Fused Deposition Modeling (FDM)** technique for prototyping. It is simple, cost-effective, and ideal for producing strong, lightweight parts suitable for testing and everyday use.

4.1 Pre-Processing

In this stage, the CAD model was exported into STL format and opened in slicing software. The software was used to set parameters like layer height, infill density, and print orientation. This step ensured that the model was optimized for efficient printing with minimum errors.





4.2 Post-Processing

After printing, the prototype was carefully removed from the printer bed. Support material, if any, was cleaned, and the surface was slightly finished to improve the appearance. This step made the prototype ready for use and testing.

5. Different Materials of 3D Printing Machines

1. **PLA (Polylactic Acid):**
   A biodegradable plastic made from natural resources like corn starch. It is easy to print, environmentally friendly, and commonly used for prototypes and models.

2. **ABS (Acrylonitrile Butadiene Styrene):**
   A strong and durable material often used for functional parts. It can withstand higher temperatures but requires a heated bed for better printing results.

3. **PETG (Polyethylene Terephthalate Glycol):**
   Combines the strength of ABS with the easy printing of PLA. It is durable, slightly flexible, and resistant to moisture, making it suitable for long-term use.

4. **Nylon:**
   A tough, flexible, and wear-resistant material. It is commonly used for mechanical parts like gears and hinges due to its high strength and durability.

5. **Resin:**
   Used mainly in SLA 3D printers, resin provides high detail and smooth surface finish. It is ideal for precise models, miniatures, and products requiring fine detailing.

6. Applications of Project and Group Assembly

Applications of the Smartphone Holder/Stand for Laptop:

1. Multitasking: Allows users to work on their laptop while keeping their smartphone in view for calls, chats, or notifications.

2. Online Learning: Students can attend online classes on one device and take notes or work on assignments on the laptop at the same time.

3. Video Conferencing: Professionals can use the holder for video calls on the phone while simultaneously working on presentations or documents on the laptop.

4. Tutorials and Coding: Users can watch tutorials or reference material on their smartphone while practicing or coding on the laptop.

5. Space Saving: Keeps the workspace neat and organized by attaching the smartphone to the laptop instead of placing it separately on the desk.

6. Portable Accessory: Lightweight and easy to carry, making it useful for students, office workers, and travelers.

Group Assembly:

The project was completed as a team activity, where each member contributed in different areas such as idea generation, CAD design, slicing and printing, and documentation. This collaborative effort made the prototype successful and ensured that the work was completed efficiently.

7. Bill of Material

Part	Qty	Material	Notes
D (Spine)	1	2 each	220 mm
C (Rails)	2	2 each	88 mm
A/B (Clamps)	2 each	2 each	Adjustable
I/J (Grips)	2 each	PLA	Top bezel hooks
E/F/G (Mobile Holder)	1 set	PLA	Adjustable jaws
H (Pivots)	4	Steel/PLA	Ø6 mm
TPU Pads	4	TPU	30×10×2 mm

8. Conclusion

Through this project, we successfully designed and developed a **Smartphone Holder/Stand for Laptop** using CAD modeling and 3D printing. The prototype demonstrated how rapid prototyping can be applied to create practical and innovative solutions for everyday problems.

This project helped us learn the complete product development process — from concept generation, CAD modeling, and slicing to 3D printing and post-processing. It also improved our teamwork, problem-solving, and technical skills.

Overall, the smartphone holder proved to be lightweight, portable, and user-friendly, making it a useful accessory for students, professionals, and anyone who multitasks with a laptop and smartphone.

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